

AHMED MUBARAK

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SUMMARY

Automation & Controls Junior Eng. at **AECOM** with hands-on experience in **PLC programming, HMI/SCADA development, electrical schematics, control panels, I/O testing, FAT/SAT, commissioning, and troubleshooting.** Experienced with **Allen-Bradley/Rockwell, Siemens, FactoryTalk View, Ignition, VFDs, sensors, field devices, control documentation, and industrial automation systems** across manufacturing, infrastructure, water/wastewater, and power generation environments. Strong early-career controls engineer with real field experience, project coordination exposure, first-principles problem solving, and a strong interest in equipment reliability, process optimization, and using AI in making process more efficient.

***** Holds Bruce Power Nuclear Clearance and Government of Canada Reliability Clearance. *****

EDUCATION

Bachelor of Automation Engineering Technology, McMaster University {Honours} *Sept 2021 - Dec 2025*
• Advanced Diploma, Chemical Engineering Process Design, Mohawk College *Sept 2021 - Dec 2023*
• Project Management Accredited Certificate, McMaster University *Sept 2021 - Dec 2025*

RELEVANT WORK EXPERIENCE

Instrumentation and Controls Engineer - EIT - AECOM *Nov 2025 - Present*

- **Lead and support controls/I&C design scopes** for multiple water/wastewater and infrastructure facilities, developing automation deliverables including **PLC/control panel design, instrumentation selection, I/O lists, SCADA/network architecture, alarm and monitoring points, equipment tagging, and detailed design submissions.**
- Review and interpret **P&IDs, process requirements, control narratives,** and existing plant control documentation to translate operational needs into practical **PLC, HMI/SCADA,** electrical, instrumentation, and control system requirements.
- Led and supported **SCADA/HMI development and upgrade projects** for water treatment and wastewater facilities using **Ignition, GE iFIX, and FactoryTalk View,** developing operator screens, tags, alarms, trends, and equipment monitoring tools that improved plant visibility, troubleshooting, and operational reliability.
- Independently led **SCADA development** for the **Doran Neighbourhood** new-build water treatment plant and wastewater pumping station using **Ignition.**
- Lead and support **FAT, SAT, commissioning, field verification, vendor package reviews, BOM development, QA/QC checks, deficiency tracking, and technical issue resolution** to validate control system performance, improve reliability, and support safe, constructible project delivery.
- Identified a major inefficiency in the proposal development workflow and took initiative to independently develop a **Python-based automation tool** that centralized project history, team resumes, workload data, and reusable technical content, reducing proposal preparation time from approximately **three weeks to three days** while improving consistency, standardization, and cost efficiency.

Automation & Controls Engineer (Co-op), JNE Group *May 2025 – Sept 2025*

- Delivered control-system design support for **US Steel's** HDGL wastewater treatment plant, including **control schematics, PLC panel layouts, 368+ point I/O list development, FAT/SAT support,** and on-site commissioning.
- Programmed, tested, and supported **Siemens PLC, Allen-Bradley PLC, FactoryTalk View HMI,** and **SCADA/HMI** applications for industrial automation systems, including equipment status, alarms, operator controls, process monitoring, and troubleshooting support.
- Developed and reviewed **low-voltage industrial control system** deliverables, including **control schematics, panel layouts, single-line diagrams, load lists, termination details, instrument connections, installation details, and I/O documentation.**
- Supported vendor package reviews, equipment selection, BOM development, network architecture review, and control panel documentation with exposure to **CSA, CEC, UL/UL 508A, and Class I Div 2 hazardous-location design considerations.**

Control Systems Engineer, Midwest Services Canada

Nov 2024 – May 2025

- Designed and implemented **Siemens PLC-based automation** and **low-voltage control systems** for commercial and industrial building applications, including HVAC, compressor controls, alarms, equipment monitoring, and operator interfaces.
- **Provided full design and implementation** for the *Delta Marriott Hotel* networked dual-compressor control system, including **electrical design, PLC programming, control panel design, device selection, commissioning, and operator training**.
- Developed electrical schematics, control circuits, panel layouts, wiring diagrams, termination details, load requirements, BOMs, and installation documentation for fabrication, certification, and installation.
- Selected and integrated control panel components with consideration for **CSA, CEC, and ESA/eSAFE**, including PLCs, power supplies, relays, terminal blocks, protection devices, switches, indicators, and communication hardware.

Electrical/Controls Field Engineer (Co-op), GE Vernova (General Electric)

Jan 2024 - Sept 2024

- Supported commissioning, maintenance, troubleshooting, and upgrades of DCS and electrical control systems for **gas turbines, steam turbines, and generators** during **Ontario's 2024 spring outage season** across **five power stations**.
- Performed field commissioning, loop checks, signal verification, alarm review, instrumentation troubleshooting, and control system validation for critical power generation assets.
- Reviewed electrical drawings, control schematics, wiring diagrams, termination information, field device connections, vendor documentation, and commissioning records to support testing.
- Supported **HMI/SCADA** and **DCS troubleshooting** by reviewing alarms, field signals, control logic, operator interface issues, and equipment status to improve system reliability during outage execution.

SKILLS

PLC / HMI / SCADA Platforms: Allen-Bradley/Rockwell Studio 5000, RSLogix 5000, Siemens TIA Portal, Siemens LOGO!, FactoryTalk View, Ignition, GE iFIX / Proficy / Cimplicity, Siemens WinCC, and AVEVA
Networks / Industrial Expertise: Industrial communication protocols (EtherNet/IP, PROFINET, Modbus TCP/IP, BACnet, HART), instrumentation sizing, FAT/SAT, commissioning, QA/QC, CSA & ISA standards, VFDs, AC drive control, motor control, sensors, actuators.

Programming & Documentation: Python, technical documentation, AutoCAD, Revit, BOM development

ACADEMIC EXPERIENCE

Automation LAB Research Assistant (Part-Time) McMaster University

Sept 2024 – May 2025

- Designing and building an RC car platform with advanced control systems.
- Programmed and wired a conveyor line using **Allen-Bradley PLC** and **HMI** using **Factory Talk View**.
- Configured and tuned (**VFDs**) for precise motor control in a conveyor system.

Allen Bradley PLC Advanced (3PL3), TA, McMaster University

Sept 2025 - Dec 2025

Allen Bradley PLC (2PL3), TA, McMaster University

Jan 2025 - April 2025

Chemical Engineering Process Design (2EC3), TA, McMaster University

Jan 2025 - April 2025

Math 1 (1MC3), TA, McMaster University

Sept 2024 - Dec 2024

SAMPLE PERSONAL PROJECT

HVAC Air Compressors Control System – Delta Ottawa Hotel (Capstone)

Jan 2025 – March 2025

- **Led the end-to-end design and implementation** of a dual air compressor building automation system powered by a **Siemens LOGO! PLC** and **CSA-compliant custom panels**.
- **Designed full system architecture**, Electrical schematics, power distribution, control circuits, and protection schemes for fabrication and **ESA/ESAFE** certification.
- **Programmed control logic** for automatic/manual operation, load balancing, runtime tracking, and maintenance scheduling, extending compressor lifecycle and reliability.
- Integrated advanced alarm handling (leak detection, low pressure, abnormal operation) and automated daily engineer reports, providing operators with real-time visibility of system health.
- **Managed project planning**, supplier coordination, commissioning, and engineer training, delivering a fully certified, standards-compliant solution that reduced downtime and improved operational safety.